CANONICAL METRICS ON SINGULAR MODULI OF K3 BUNDLES

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Abstract. The canonical singular Ricci-flat metric on a class of compact K3 bundle moduli spaces admits quadratic approximation by its hyperkähler tangent cones in holomorphic coordinates. The spaces parametrize twice a primitive Mukai vector on a K3 surface of Picard rank one; a rank bound ensures that every semistable sheaf and every slope Jordan–Hölder factor is locally free. These spaces have positive-dimensional singular strata and admit neither a symplectic resolution nor a smoothing. We identify the metric in an ample determinant class with the harmonic L^2 metric, identify its tangent cones as the flat hyperkähler quotients of the extension spaces, and prove that its regular-locus completion is the original moduli space. The comparison is uniform through changes of stabilizer in each fixed local chart. The proof combines quantitative local models for Hermitian–Einstein metrics with exact quadraticity in harmonic Kuranishi coordinates, continuous Weil–Petersson potentials, and a determinant-class calculation.

1. Introduction

A projective symplectic variety carries a unique singular Ricci-flat Kähler metric in each Kähler class, in the sense of Eyssidieux–Guedj–Zeriahi [EGZ09, Theorem 7.5 and Corollary 7.6]. The existence theorem does not by itself describe the local metric geometry at a singular point. In a moduli space of bundles, a second description is available: Hermitian–Einstein connections define a hyperkähler L^2 metric on the stable locus. We identify these two metrics on a family of singular compact K3 moduli spaces and obtain a quadratic comparison with explicit hyperkähler cones at every singular point.

The simplest example is already a compact space with two positive-dimensional singular strata. Let S be a quartic K3 surface with $\mathrm{Pic}(S) = \mathbb{Z}H$, $H^2 = 4$, and put

$$v_0 = (5, H, 0), \qquad v = 2v_0 = (10, 2H, 0), \qquad M = M_H(v). \tag{1}$$

Here $M_H(v)$ denotes the moduli space of H-Gieseker-semistable sheaves of Mukai vector v. This space has complex dimension 18. Every semistable sheaf is a vector bundle, and its slope Jordan–Hölder factors are vector bundles as well. If $N = M_H(v_0)$, then N is a smooth compact hyperkähler sixfold, and the singular locus and its diagonal are

$$\Sigma = \mathrm{Sym}^2 N, \qquad \Delta = \{[G \oplus G] : [G] \in N\}. \tag{2}$$

Thus $\dim_{\mathbb{C}} \Sigma = 12$ and $\dim_{\mathbb{C}} \Delta = 6$. The variety M has terminal, locally factorial symplectic singularities, admits no symplectic resolution, and has no smoothing. These assertions follow from the moduli results recalled in Section 2. The exclusion of a non-locally-free boundary is proved there by a Mukai-pairing inequality.

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1.1. **The family and the main result.** Let S be a projective K3 surface with $\mathrm{Pic}(S) = \mathbb{Z}H$, where H is ample. Write the Mukai vector of a sheaf as $v(F) = (\mathrm{rk}\, F, c_1(F), \chi(F) - \mathrm{rk}\, F)$, with pairing

$$\langle (r, c, s), (r', c', s') \rangle = c \cdot c' - rs' - r's.$$

We impose the conditions

$$v_0 = (r_0, aH, s_0), \qquad \gcd(r_0, aH^2) = 1, \qquad q := v_0^2 \geq 4, \qquad r_0 \geq q + 1, \tag{3}$$

and set

$$M = M_H(2v_0), \qquad N = M_H(v_0). \tag{4}$$

The integer q is even. Fix the Ricci-flat Kähler form $\omega \in c_1(H)$ on S. Proposition 2.1 proves that slope and Gieseker semistability agree for the sheaves under consideration. The moduli results in Section 2 identify the stable locus M^s with the regular locus of M. The Hermitian–Einstein metrics therefore define the harmonic L^2 metric g^{L^2} on M_{reg}.

Here is the local cone that occurs in the theorem. For a polystable bundle E, let

$$V_E = H^1(S, \mathrm{End}\, E), \qquad K_E = \mathrm{Aut}(E, h_E), \qquad \mathcal{C}_E = V_E /\!\!/\!\!/_0\, K_E. \tag{5}$$

Harmonic representatives give V_E its flat hyperkähler structure. The last expression means the simultaneous zero set of the three moment maps, divided by the unitary automorphism group; an ineffective scalar circle can be removed. The regular part of this quotient carries the reduction metric $g_{\mathcal{C}_E}$. Its full length metric is denoted by $d_{\mathcal{C}_E}$, and $r(x) = d_{\mathcal{C}_E}(0, x)$. Equivalently, $\mathcal{C}_E$ is the flat hyperkähler quotient of the doubled Ext-quiver of the stable summands of E. Section 3 specifies the harmonic and moment-map conventions.

Theorem 1.1. *Assume (3). There is an ample Donaldson determinant line bundle λ on M and a constant $\kappa > 0$ with the following properties.*

1. *The Kähler form of g^{L^2} extends to the canonical singular Ricci-flat current in the class $\kappa c_1(\lambda)$. It has continuous local potentials on M.*
2. *At every polystable point $[E] \in M$ there is a biholomorphism of germs*

$$\Psi_E : (\mathcal{C}_E, 0) \longrightarrow (M, [E])$$

for which, on the regular locus of a sufficiently small neighbourhood,

$$(1 - C_E r^2) g_{\mathcal{C}_E} \leq \Psi_E^* g^{L^2} \leq (1 + C_E r^2) g_{\mathcal{C}_E}. \tag{6}$$

The same inequality holds on tangent vectors to every smooth orbit-type stratum, with its harmonic metric. Within this fixed chart, C_E is independent of stabilizer type and of the ratios of positive orbit eigenvalues.

The determinant convention, including its positive sign, is given in Section 5. The constants and chart radii in (6) may depend on the fixed bundle E, its harmonic dimensions and its positive background elliptic gaps. The theorem does not assert one radius or comparison constant for all possible centred charts.

Corollary 1.2. *The metric completion of $(M_{\mathrm{reg}}, g^{L^2})$ is a compact geodesic space canonically homeomorphic to M. At every $p = [E] \in M$ its metric tangent cone is unique and is $(\mathcal{C}_E, d_{\mathcal{C}_E}, 0)$. In the chart of Theorem 1.1,*

$$d(p, \Psi_E(x)) = r(x)(1 + O_E(r(x)^2)). \tag{7}$$

The Gromov–Hausdorff distance between the closed unit balls in $(M, \epsilon^{-1} d, p)$ and in $\mathcal{C}_E$ is $O_E(\epsilon^2)$.

The completion assertion requires a path argument in addition to the tensor estimate. We prove that paths along singular strata can be approximated in length by regular paths. The perturbation is transverse to the stratum in a product slice; radial dilation alone preserves the singular stratum.

In the example (1), the two singular tangent cones are

$$\begin{aligned} \mathcal{C}_{G_1\oplus G_2} &\cong \mathbb{C}^{12}\times (T^*\mathbb{C}^4 /\!\!/\!\!/_0\, \mathrm{U}(1)), \qquad & G_1 \ncong G_2,\\ \mathcal{C}_{G\oplus G} &\cong \mathbb{C}^{6}\times ((\mathbb{C}^6\otimes \mathfrak{sl}_2) /\!\!/\!\!/_0\, \mathrm{SU}(2)). \end{aligned} \tag{8}$$

The first normal factor is the closure of the minimal nilpotent orbit in $\mathfrak{sl}_4$, of complex dimension six. The second normal factor has complex dimension twelve. These products are metric products, with the flat factors normalized by the harmonic inner products.

The approximation in (6) is not an exact local isometry. Proposition 6.3 proves that no singular point has a neighbourhood isometric to a neighbourhood of the vertex in its tangent cone. The reason is the intrinsic geometry of the singular strata: their induced length metrics contain the nonflat hyperkähler metric of N. This does not assert that the exponent two is optimal.

1.2. **Relation to earlier work.** The algebraic local models for these singularities belong to the deformation theory of sheaves on K3 surfaces. Kaledin–Lehn–Sorger established the structure and nonexistence of symplectic resolutions for the relevant nonprimitive moduli spaces [KLS06]. Arbarello–Saccà identified the local quiver germs under a formality hypothesis [AS18, Propositions 4.4 and 6.1], and Budur–Zhang proved that hypothesis for polystable sheaves on projective K3 surfaces [BZ19, Theorem 1.2]. Perego–Rapagnetta established the irreducible symplectic-variety properties used here [PR23].

On the analytic side, Itoh proved that the L^2 metric on the irreducible Einstein–Hermitian moduli space over a K3 surface is hyperkähler [Ito88, Theorem 2.1 and Remark (iii)]. Verbitsky's theory of hyperholomorphic and projectively hyperholomorphic bundles gives the underlying quadraticity mechanism [Ver96, Theorems 6.2, 7.1 and 11.2, arXiv version]. We use a surface calculation to show that the obstruction map is quadratic in the particular harmonic coordinates needed for the metric estimate. An unspecified analytic change of coordinates would not preserve the stated quadratic error.

The quantitative local comparison for the harmonic metric is imported from [Wue26b, Theorem 1.1(iii)]. Its application here uses the full harmonic Kuranishi locus, including repeated stable factors. The finite-dimensional metric slices and localization arguments are those of [Wue26a, Sections 4–6]. We state the required versions below. The additional global step uses continuous Weil–Petersson potentials from Buchdahl–Schumacher [BSc21, Theorem 11], the curvature formula of Biswas–Schumacher [BS16, Proposition 4.3], and uniqueness of the singular Ricci-flat current. Continuous Weil–Petersson potentials themselves are an established input.

Namikawa–Odaka study canonical metrics and conical structures on symplectic singularities, including a theorem under a symplectic resolution or smoothing hypothesis and higher-order asymptotics at isolated singularities [NO26, Theorem 1.1 and Corollary 6.8]. Odaka's subsequent result gives a hyperkähler cone structure on every symplectic singularity germ [Oda26, Corollary 1.3]. The distinction relevant here is between the existence of a local cone metric and the asymptotics of the metric selected by an ample class on a compact moduli space. Theorem 1.1 concerns the latter metric and gives a relative estimate across its positive-dimensional singular strata. It makes no assertion about higher derivatives near lower strata or about the complete Donaldson–Sun degeneration data.

1.3. **Proof and organization.** The local argument begins with the canonical harmonic reconstruction $a = x - \bar\partial^* G(a \wedge a)$. On a Ricci-flat K3 surface, harmonic endomorphism-valued two-forms have parallel coefficients. Integration by parts then annihilates the harmonic projection of every product containing the coexact correction. Thus the obstruction is exactly quadratic in the coordinates in which the Hermitian–Einstein comparison has quadratic error. The holomorphic family and the metric reconstruction classify the same complex gauge orbit, so the resulting chart is simultaneously holomorphic and quantitative. Transverse perturbations in product slices then identify the full stratified length distance with the completion of the regular-locus metric.

For the global identification, the rank inequality forces the defect cycles of all slope graded objects to vanish. The Donaldson–Uhlenbeck morphism is consequently finite, making its determinant polarization ample. A discriminant calculation identifies this class with the harmonic Kähler class. Continuous local potentials, together with a depth-based cohomology restriction theorem, produce a global continuous potential in that class. The hyperkähler volume identity then places the resulting current in the uniqueness theorem of Eyssidieux–Guedj–Zeriahi.

Section 2 proves that the rank bound excludes both non-locally-free sheaves and non-locally-free graded factors, and describes the two singular strata. Section 3 proves exact harmonic quadraticity and applies the quantitative Hermitian–Einstein comparison in the same holomorphic quotient coordinates. Section 4 identifies the completion distance and derives the rescaling estimates. Section 5 identifies the ample determinant class and proves the global Ricci-flat metric statement. Section 6 proves the failure of local isometry.

All dimensions below are complex unless otherwise indicated. We use $d^c = \frac{i}{2}(\bar\partial - \partial)$, so $dd^c = i\partial\bar\partial$. A smooth stratum carries its harmonic L^2 metric; a distance on the entire stratified space will always be specified by its allowed paths.

2. A compact moduli space of bundles

We first show that the rank condition in (3) excludes both singular sheaves and singular stable factors. The second assertion is needed for the determinant polarization: the cycle recorded by the Donaldson–Uhlenbeck morphism is the defect of the slope graded object, not just the defect of the original sheaf.

Proposition 2.1. *Assume (3). Every slope-semistable torsion-free sheaf of Mukai vector $2v_0$ is locally free. If it is strictly slope-semistable, its slope Jordan–Hölder factors are two stable bundles of vector v_0. Every slope-semistable sheaf of vector v_0 is stable and locally free. For these two Mukai vectors, slope and Gieseker semistability agree, as do stability and S-equivalence.*

Proof. We use two elementary properties of stable sheaves on a K3 surface. If A is stable, simplicity and Serre duality give

$$v(A)^2 = \dim \operatorname{Ext}^1(A, A) - 2 \geq -2. \tag{9}$$

If A is slope-stable and torsion-free, A^{**} is a stable bundle, and

$$v(A^{**}) = v(A) + (0, 0, \operatorname{length}(A^{**}/A)).$$

The reflexive hull of a slope-semistable sheaf is semistable.

Let F have vector $2v_0$, and write $\ell = \operatorname{length}(F^{**}/F)$. If F^{**} is stable, (9) implies

$$4q - 4r_0\ell \geq -2.$$

Since $r_0 \geq q+1$, this forces $\ell = 0$.

Suppose instead that F^{**} is strictly slope-semistable. An equal-slope factor of rank r' and first Chern class $a'H$ satisfies

$$r_0 a' H^2 = r' a H^2.$$

Coprimality in (3) forces $r_0 \mid r'$. There are therefore exactly two stable factors A_1, A_2, each of rank r_0 and first Chern class aH. Put

$$v(A_i) = (r_0, aH, s_i), \qquad \ell_i = \operatorname{length}(A_i^{**}/A_i).$$

Additivity gives $s_1 + s_2 = 2s_0 + \ell$. Applied to A_i^{**}, (9) gives

$$q - 2r_0(s_i + \ell_i - s_0) \geq -2. \tag{10}$$

Because $2r_0 > q+2$, integrality implies $s_i + \ell_i \leq s_0$. Summing yields $\ell + \ell_1 + \ell_2 \leq 0$. All three lengths vanish, and $s_1 = s_2 = s_0$. Thus F is a bundle and its stable factors are bundles of vector v_0. Extensions of locally free sheaves are locally free, consistently with this conclusion.

For vector v_0, the same coprimality rules out an equal-slope proper factor. If A is such a stable sheaf, its reflexive hull has square $q - 2r_0\ell$, and (9) again forces $\ell = 0$.

Slope stability implies Gieseker stability, and Gieseker semistability implies slope semistability. A strictly slope-semistable F of vector $2v_0$ has the two factors just described. Their reduced Hilbert polynomials both equal that of F, so F is Gieseker-semistable and is not Gieseker-stable. For vector v_0 there is no strictly slope-semistable case. The Jordan–Hölder factors in the two theories consequently coincide, proving the S-equivalence assertion. □

2.1. **Geometry of the moduli space.** The vector v_0 is primitive. On a surface of Picard rank one, H is general in the sense required for these moduli spaces [PR23, Definition 2.8 and Lemma 2.9(1)]. The primitive space N is a nonempty smooth projective irreducible holomorphic symplectic manifold of dimension

$$n = q + 2, \tag{11}$$

deformation equivalent to $S^{[q/2+1]}$ [PR23, Theorem 1.7 and the paragraph following Definition 1.6]. In particular, it is simply connected. Proposition 2.1 identifies it with a compact moduli space of slope-stable bundles.

The results of Kaledin–Lehn–Sorger [KLS06, Theorem 4.4, Proposition 6.1 and Theorems A–B] give the following description. The variety M is normal, irreducible and projective, of dimension $4q+2$. Its regular locus is precisely the nonempty stable locus. Every singular point is represented by $G_1 \oplus G_2$ with $G_i \in N$. The direct-sum morphism identifies the reduced singular locus with $\operatorname{Sym}^2 N$. One can also see the last assertion from the local quiver descriptions below: the semisimple nonsimple representations are exactly the unordered pairs of simple summands. Near a repeated summand, the restrictions of traces of matrix words to two one-dimensional characters are the polarized power sums. These generate the symmetric-square invariant ring in characteristic zero, identifying the reduced singular locus. Away from the diagonal the local product description gives the same identification. Accordingly,

$$\operatorname{Sing} M = \Sigma = \operatorname{Sym}^2 N, \qquad \Delta \cong N, \qquad \dim \Sigma = 2q+4, \quad \dim \Delta = q+2. \tag{12}$$

There are three smooth strata: M^s, $\Sigma \setminus \Delta$, and Δ.

The variety M has symplectic singularities and is locally factorial in the algebraic sense. Its singular locus has codimension $2q-2 \geq 6$, so its singularities are terminal [Nam01, Corollary 1]. It has no proper symplectic resolution [KLS06, Theorem B]. The Main Theorem of [Nam06] applies to the projective, $\mathbb{Q}$-factorial, terminal symplectic variety M and makes

every flat deformation locally trivial. Since Σ is nonempty, M has no flat smoothing. Algebraic local factoriality here does not assert factoriality of completed local rings; see [KLS06, Remark 6.3].

We shall also use the irreducible symplectic-variety theorem of Perego–Rapagnetta [PR23, Theorem 1.10]. It gives canonical singularities and

$$H^0(M,\Omega_M^{[2]}) = \mathbb{C}\sigma_M, \qquad H^1(M,\mathcal{O}_M) = 0. \tag{13}$$

For the second equality, the definition of an irreducible symplectic variety gives no reflexive one-forms. Pullback to a projective resolution, Hodge symmetry, and rationality of canonical singularities give the stated cohomology vanishing; compare [PR23, Definition 1.2 and Remark 1.3]. Symplectic singularities are Gorenstein, with trivial canonical bundle generated by the top exterior power of σ_M.

2.2. **The two extension cones.** The formulas below use the harmonic Hermitian forms; the displayed complex vector spaces are supplied with the corresponding flat hyperkähler structures. A quiver representation is called simple if it has no nonzero proper subrepresentation. Let $G_1 \ncong G_2$ belong to N. Stability and Serre duality give

$$\operatorname{Hom}(G_1,G_2) = \operatorname{Ext}^2(G_1,G_2) = 0, \qquad \dim \operatorname{Ext}^1(G_1,G_2) = q.$$

Each self-extension space has dimension n. The effective unitary stabilizer is a circle, acting with opposite weights on the two cross-extension spaces. Hence

$$\mathcal{C}_{G_1\oplus G_2} = \mathbb{C}^{2n} \times \mathcal{N}_q, \qquad \mathcal{N}_q = T^*\mathbb{C}^q /\!\!/\!\!/_0 \,\mathrm{U}(1). \tag{14}$$

In coordinates $(x,y) \in \mathbb{C}^q \oplus (\mathbb{C}^q)^*$ the equations are $y(x) = 0$ and $|x|^2 = |y|^2$. The map $(x,y) \mapsto x \otimes y$ identifies the complex quotient with the rank-at-most-one, square-zero matrices in $\mathfrak{sl}_q$, namely the minimal nilpotent orbit closure. Its complex dimension is $2q-2$. There are simple balanced points: take nonzero perpendicular vectors of equal norm for x and the conjugate of y.

For $E = G \oplus G$, the extension space is $\operatorname{Ext}^1(G,G) \otimes \mathfrak{gl}_2$. Splitting off scalar matrices gives

$$\mathcal{C}_{G\oplus G} = \mathbb{C}^n \times \mathcal{Q}_q, \qquad \mathcal{Q}_q = (\mathbb{C}^n \otimes \mathfrak{sl}_2) /\!\!/\!\!/_0 \,\mathrm{SU}(2), \qquad \dim \mathcal{Q}_q = 3q. \tag{15}$$

Here n is even, and a symplectic basis identifies the extension space with $n/2$ pairs of matrix loops (A_j, B_j). Simple balanced points exist: set all $B_j = 0$ and choose two noncommuting Hermitian traceless matrices among the A_j, with the remaining matrices zero. The complex moment map $\sum_j [A_j, B_j]$ and the real moment map $\sum_j ([A_j, A_j^*] + [B_j, B_j^*])$ vanish. The two Hermitian matrices have no common invariant line, so this representation is simple. Scaling makes these simple points arbitrarily small. The scalar center acts trivially, so the use of $\mathrm{SU}(2)$ rather than its effective quotient does not change the metric quotient.

The holomorphic germs and metric normalizations in (14)–(15) will be identified with those of M in the next section. In particular, the metric on the fixed scalar factor of (15) is the one induced by $a \mapsto a \otimes I_2$; it is twice the harmonic metric on the deformation space of G.

Example 2.2. For a Picard-rank-one quartic K3 surface, $v_0 = (5, H, 0)$ satisfies $q = 4$, $r_0 = q+1$ and $\gcd(5,4) = 1$. The stable factors have second Chern class $c_2 = 7$, and a sheaf of vector $2v_0$ has $c_2 = 18$, since $s = r + c_1^2/2 - c_2$ on a K3 surface. The dimensions are as follows.

Locus	Complex dimension	Complex normal-cone dimension
M^s	18	0
$\Sigma \setminus \Delta$	12	6
Δ	6	12

Thus the family contains a compact example with both types of singular points. The existence assertions are the moduli theorems above; the numerical equalities alone do not establish existence.

3. Harmonic coordinates and the quadratic estimate

Let $E = \bigoplus_i F_i \otimes W_i$ be a polystable bundle on S, with pairwise nonisomorphic stable factors and a direct-sum Hermitian–Einstein metric h_E. Put

$$V = H^1(S, \operatorname{End} E), \qquad G = \prod_i \mathrm{GL}(W_i), \qquad K = \prod_i \mathrm{U}(W_i).$$

We use unnormalized trace inner products and volume form $\omega^2/2$. Write $d = \bar\partial_E$, and let H_j, G_j denote harmonic projection and the Green operator in degree $(0, j)$. The flat metric on V is the real metric of this harmonic Hermitian form. Its quaternionic structure is induced by the parallel holomorphic symplectic form of S, normalized to make that structure antiunitary. All quotient metrics use these same normalizations.

3.1. **Exact quadraticity in the harmonic coordinates.** The canonical harmonic recursion is

$$a(x) = x - d^* G_2(a(x) \wedge a(x)), \qquad \kappa_E(x) = H_2(a(x) \wedge a(x)). \tag{16}$$

It converges on a neighbourhood of $0 \in V$ and gives a holomorphic semiuniversal deformation over $Z = \kappa_E^{-1}(0)$; its underlying reduced space suffices for all metric statements. In particular,

$$H_1 a = x, \qquad d^* a = 0, \qquad a = x + O(|x|^2). \tag{17}$$

The maps are locally G-equivariant. These assertions, including integrability precisely on Z, are the harmonic construction of [Wue26b, Lemma 2.1]. Convergence follows from the contraction for $d^* G_2(a \wedge a)$ in a sufficiently high Sobolev norm. Parallel endomorphisms commute with the Hodge operators, which gives the equivariance even for nonunitary elements of G whenever both sides are defined.

Lemma 3.1. *On the full convergence neighbourhood of* (16),

$$\kappa_E(x) = H_2(x \wedge x). \tag{18}$$

Proof. Choose a nonzero parallel holomorphic two-form σ on S. Every endomorphism-valued $(0,2)$-form is $\phi\bar\sigma$. The formula for d^*, together with parallelness of σ, identifies $d^*(\phi\bar\sigma)$ with a nonzero constant multiple of contraction of $\partial_E\phi$ by $\bar\sigma$. Thus $\phi\bar\sigma$ is harmonic if and only if $\partial_E\phi = 0$. Equivalently, ϕ^* is holomorphic. The induced connection on $\operatorname{End} E$ has zero Hermitian–Einstein constant, so its holomorphic sections are parallel. We obtain

$$\mathcal{H}^{0,2}(\operatorname{End} E) = \{\xi\bar\sigma : \nabla\xi = 0\}. \tag{19}$$

Consequently, for $\theta = \theta_0\bar\sigma$,

$$H_2\theta = 0 \quad \Longleftrightarrow \quad \int_S \operatorname{tr}(\theta_0 \xi^*)\, dV = 0 \quad \text{for every parallel } \xi. \tag{20}$$

We claim that for $d^*u = 0$ and arbitrary $\gamma = \phi\bar{\sigma}$,

$$H_2(u \wedge d^*\gamma) = 0 = H_2(d^*\gamma \wedge u). \tag{21}$$

Use unitary normal frames at a point. Up to fixed nonzero scalars, the two coefficients relative to $\bar{\sigma}$ are

$$-\sum_j u_{\bar{j}} \nabla_j \phi, \qquad \sum_j (\nabla_j \phi) u_{\bar{j}}.$$

For a parallel ξ, covariant integration by parts gives

$$\int_S \sum_j \operatorname{tr}(u_{\bar{j}} \nabla_j \phi\, \xi^*)\, dV = -\int_S \operatorname{tr}\left(\left(\sum_j \nabla_j u_{\bar{j}}\right) \phi \xi^*\right) dV = 0, \tag{22}$$

$$\int_S \sum_j \operatorname{tr}((\nabla_j \phi) u_{\bar{j}} \xi^*)\, dV = -\int_S \operatorname{tr}\left(\phi \left(\sum_j \nabla_j u_{\bar{j}}\right) \xi^*\right) dV = 0. \tag{23}$$

The contracted derivative is a constant multiple of $-d^*u$. The connection preserves the metric and volume, and $\nabla\xi^* = 0$, so these are global divergence identities. The order of the endomorphisms is retained in both formulas; neither commuting factors nor flatness of the Chern connection is used. Equations (20)–(23) prove (21).

Now write $a = x + \beta$, where $\beta = -d^*G_2(a \wedge a)$. Both x and β are d^*-closed. Formula (21), first with $u = x$ and then with $u = \beta$, annihilates under H_2 all three terms $x \wedge \beta$, $\beta \wedge x$ and $\beta \wedge \beta$. Expanding $H_2(a \wedge a)$ proves (18). □

This calculation specifies the coordinates in the quadraticity theorems for hyperholomorphic bundles. It uses the central Ricci-flat metric on the surface. The coordinate specification is essential for the metric estimate below: a general holomorphic substitution $x \mapsto x + O(|x|^2)$ has derivative $I + O(|x|)$ and need only preserve a relative linear error.

Choose unitary harmonic bases for the cross-extension spaces and symplectically paired bases for the loops. Serre duality identifies $H_2(x \wedge x)$ with the complex moment map of this doubled quiver. More precisely, with $r_i = \operatorname{rk} F_i$ and a normalized harmonic generator η_S of $H^2(S, \mathcal{O}_S)$,

$$H_2(x \wedge x) = \bigoplus_i r_i^{-1} I_{F_i} \otimes \mu_{\mathbb{C},i}(x)\, \eta_S, \tag{24}$$

up to the single common nonzero symplectic normalization; see the rank-dual formula in [Wue26a, Section 3]. In particular the zero sets agree exactly. Since $H^1(S, \mathcal{O}_S) = 0$, fixing the determinant removes no harmonic direction. Lemma 3.1 therefore gives

$$Z = \mu_{\mathbb{C}}^{-1}(0), \qquad (Z \cap \mu_{\mathbb{R}}^{-1}(0))/K = \mathcal{C}_E. \tag{25}$$

3.2. **The quantitative local comparison.** We record precisely the metric input from [Wue26b, Theorem 1.1(i),(iii)], at constant polarization. For a fixed locally free polystable background on a compact Kähler manifold, in the coordinates (16) and on the fixed-determinant subspace, that theorem gives a homeomorphism

$$f_0 : (Z \cap \mu_{\mathbb{R}}^{-1}(0))/K \longrightarrow \text{a neighbourhood of } [E]$$

onto the nearby Hermitian–Einstein moduli. It covers every sufficiently nearby solution, preserves the local complex-orbit description, and is smooth on smooth pieces of fixed orbit type. For genuine tangent vectors to such pieces,

$$(1 - C|x|^2) g_{\text{fin}} \le f_0^* g^{L^2} \le (1 + C|x|^2) g_{\text{fin}}. \tag{26}$$

Here, if v is a balanced tangent lift and $C_x\xi = [\xi, x]$,

$$g_{\mathrm{fin},x}([v],[v]) = \inf_{\xi} \|v - C_x\xi\|^2$$

is the flat reduction metric; the scalar kernel can be omitted from the gauge algebra. This is a statement on actual stratum tangents, not on all Zariski tangents at a singular point. The constant is independent of stabilizer type, of the positive singular values of C_x or $D\kappa_E(x)$, and of ratios of positive orbit eigenvalues. It may depend on the fixed background, ranks and multiplicities, harmonic dimensions, positive elliptic gaps, Sobolev constants and the chosen regularity order. These are exactly the dependencies needed in (6).

All hypotheses hold here by Proposition 2.1. The Hermitian–Einstein correspondence identifies the analytic moduli with the slope-polystable bundle moduli, and the latter is the Gieseker moduli germ by that proposition. By (25), $g_{\mathrm{fin}} = g_{\mathcal{C}_E}$. A zero-level balanced representative satisfies

$$r([x]) = \|x\|. \tag{27}$$

Indeed, the radial segment is horizontal and has length $\|x\|$, while the norm is 1-Lipschitz on every stratum. The full cone distance has the same property [Wue26a, Section 5]. Thus (26) is precisely the asserted quadratic tensor estimate.

It remains to check that the identification is holomorphic in these same coordinates. The family $\bar\partial_E + a(x)$ is holomorphic, locally G-equivariant and semiuniversal. The analytic GIT construction of Buchdahl–Schumacher [BSc21, Theorems 5–9 and Remark 4] identifies its reduced analytic quotient $Z//G$ with the reduced moduli germ. Denote its holomorphic classifying isomorphism by χ. The Kempf–Ness identification supplies the homeomorphism between the balanced quotient in (25) and $Z//G$.

The map f_0 in (26) corrects the same operator $\bar\partial_E + a(x)$ within its complex gauge orbit. Its finite balancing step changes x within its G-orbit. Consequently f_0 and χ assign the same bundle class to every balanced representative. Local coverage and uniqueness identify them. The holomorphic structure on the cone quotient therefore makes $\Psi_E = f_0$ a biholomorphism of germs, without any further change of coordinates. This proves part (ii) of Theorem 1.1.

3.3. **Finite-dimensional slices used below.** We also need two finite-dimensional conclusions of [Wue26a, Sections 4–6]. At a fixed point of a flat zero-level doubled-quiver quotient, its full quaternionic normal representation W is again a doubled Ext-quiver representation. If K' is the stabilizer and $U = W^{K'}$, the normal quotient splits as the exact metric product

$$W /\!\!/\!\!/_0 K' = U \times \left(U^{\perp} /\!\!/\!\!/_0 K'\right). \tag{28}$$

The stratum through the centre is $U \times \{0\}$. There is a slice homeomorphism, smooth on strata, with relative metric error $O(\|w\|)$ and derivative at the central stratum equal to the isometric normal identification. Its constant may depend on that centre, but not on positive eigenvalues of the retained orbit operator.

The quotient is proper and geodesic, with its quotient topology. If its complex zero fibre has one module-simple point, then the simple locus is dense and every full normal Ext-quiver has a module-simple point as well. These are the metric-slice theorem, the properness theorem and the normal-simple lemma of [Wue26a]. The explicit simple points in Section 2.2 verify this hypothesis for our two singular cones. These statements apply to the full W in (28); the complementary representation $U^{\perp}$ alone need not be a doubled quiver.

4. Completion and tangent cones

We distinguish two path distances until their equality has been proved. On M_{reg}, let d_{reg} be the Riemannian distance of g^{L^2}. On all of M, call a path admissible if it is continuous, has a finite partition into smooth paths lying in single strata on the open subintervals, and has finite one-sided stratum lengths at the endpoints. Let d_{str} be the infimum of these lengths. The same convention on each finite cone gives its full length distance $d_{\mathcal{C}_E}$. Piecewise smooth paths and their finite-length limits give the same infimum.

4.1. **Short paths in the regular locus.**

The following elementary consequence of semialgebraicity provides control near the endpoints of transverse perturbations.

Lemma 4.1. *Let $\mathcal{C}$ be one of the cones in (14)–(15), or a full normal cone to one of its strata. There is a constant $A_{\mathcal{C}}$ such that any two regular points of radius at most ρ can be joined in the regular locus, within that radial ball, by a piecewise smooth path of length at most $A_{\mathcal{C}}\rho$.*

Proof. Write the full normal representation as W with compact group K', and let B^* be the module-simple part of the simultaneous zero set of its real and complex moment maps. The simple locus of $\mathcal{C}$ is a connected complex open subset of an irreducible cone. For the two original cones, irreducibility and normality follow from their moduli germs; homogeneity extends these properties to the whole cone. The same argument applies to full normal slices, using the slice theorem. Simplicity agrees with regularity here by the stable-locus description of Section 2 and its slice restriction. Existence and density of simple points in every normal cone were recalled in Section 3.3.

The map $B^* \to \mathcal{C}_{\mathrm{reg}}$ is proper with connected K'-orbit fibres. The group K' is a product of unitary groups, with an optional ineffective scalar quotient, and is connected. Thus B^* is connected. If the zero representation itself is simple, the effective group is trivial and the quotient is a flat vector space; the assertion is then immediate. Otherwise radial scaling identifies B^* with its unit section times $(0,\infty)$. It follows that

$$B^* \cap \{0 < \|w\| < 1\}$$

is a bounded connected semialgebraic set.

We recall why such a set has finite Euclidean inner diameter, including when it is not closed. A finite semialgebraic triangulation uses finitely many open simplices. In each simplex choose an anchor and join a variable point to that anchor along the image of the straight segment in the simplex. Join the finitely many anchors by fixed semialgebraic arcs in the connected set. This gives finitely many semialgebraic families of paths. Uniform monotonicity partitions each coordinate of each path into a bounded number of monotone intervals. Each coordinate varies over a bounded range, so its total variation, and hence the Euclidean length, has a common bound. This is the triangulation and monotonicity argument of [vD98, Sections 2 and 3, especially Theorem 3.4]. Semialgebraic arcs are piecewise smooth; endpoint derivatives may be unbounded, but their lengths are finite.

Projection from the balanced locus to its smooth quotient does not increase lengths. We obtain a uniform bound for regular points of cone radius less than one. Dilation gives the bound $A_{\mathcal{C}}\rho$ for an open radial ball. For points on its boundary, first contract both endpoints by a factor $1-\delta$, join the contracted points, and append the two regular radial segments. Their total added length is at most $2\delta\rho$. This proves the assertion for the closed ball as well, after increasing the constant. □

Proposition 4.2. *The completion of $(M_{\mathrm{reg}}, d_{\mathrm{reg}})$ is canonically homeomorphic to M, and its distance equals d_{str}. In particular it is a compact geodesic space.*

Proof. Shrink each fixed chart in Theorem 1.1(ii) so that its squared metric bounds lie between $1/2$ and 2. Lemma 4.1 then shows that the d_{reg} diameter of regular points in a radial ball of radius ρ is at most $A\rho$. Every sequence of regular points converging to the centre therefore determines the same completion point.

There is also a lower bound for leaving a chart. If an admissible path starts at cone radius ρ and first reaches the boundary of a concentric chart of radius R, its length before exit is at least

$$(R-\rho)/\sqrt{2}. \tag{29}$$

This follows by integrating the radial 1-Lipschitz inequality for the cone and the coarse tensor lower bound. In particular, distinct points of M cannot represent the same completion point. Conversely, a d_{reg}-Cauchy sequence has an algebraically convergent subsequence because M is compact. The diameter estimate identifies its limit in the completion; the Cauchy property gives the same limit for the full sequence. Thus no additional completion points occur. These estimates also prove that the completion topology is the original topology of M. Denote this completed distance by D.

We next compare lengths. Let $\gamma : [a,b] \to M$ be a C^1 path contained in a single smooth stratum. Given $\eta > 0$, cover its compact image by finitely many centred charts whose metric distortion is between $1-\eta$ and $1+\eta$. In each such chart the cone splits as $U \times \mathcal{N}$, and the stratum through its centre is $U \times \{0\}$. Subdivide $[a,b]$ so that each subarc is contained in one chart and write it there as $(u(t), 0)$.

Choose regular points $z_j \in \mathcal{N}$ tending to the vertex. They exist by the simple-point property in Section 3.3. The paths

$$t \longmapsto (u(t), z_j) \tag{30}$$

lie in the regular locus for large j. Their lengths in the exact product are the Euclidean length of u. The two tensor bounds give, for each subarc,

$$\limsup_j L_{g^{L^2}}(\Psi(u, z_j)) \le \sqrt{\frac{1+\eta}{1-\eta}}\, L_{g^{L^2}}(\gamma).$$

Their endpoints converge in the original topology, hence in D by the preceding paragraph. Taking endpoint distance limits and summing with the triangle inequality gives the same upper bound for $D(\gamma(a), \gamma(b))$. No common endpoint is required for the different perturbed subarcs. Letting η tend to zero gives $D(\gamma(a), \gamma(b)) \le L(\gamma)$.

For an admissible path whose open pieces have endpoints in lower strata, truncate each piece at both ends and apply this argument. The finite one-sided lengths and continuity of D allow the truncation parameters to tend to zero. Therefore $D \le d_{\mathrm{str}}$ on M. On regular points, every regular path is admissible, so $d_{\mathrm{str}} \le d_{\mathrm{reg}} = D$. Finally, d_{str} is continuous in the original topology: in a chart centred at any point, its radial paths have length at most $\sqrt{2}\rho$, and the triangle inequality gives continuity in both arguments. Approximation by regular endpoints extends the reverse inequality to all of M. Thus $D = d_{\mathrm{str}}$.

The completion of a length space is a length space. It is compact here, so the usual compactness argument for almost minimizing paths supplies minimizing geodesics. This proves the last assertion. □

4.2. **Quantitative convergence to the cone.** We can now transfer the quadratic tensor comparison to the completed distance. Fix a centre $p = [E]$ and its chart Ψ_E. Competitors joining points of radius at most ρ can be chosen with length $O(\rho)$, using radial paths through the centre. A competitor of this length cannot leave a fixed chart when ρ is small, by (29). Within the chart, the coarse lower bound further shows that its radius is $O(\rho)$ throughout. On all such competitors the tensor distortion is therefore $1 + O_E(\rho^2)$.

Apply this reasoning first to almost minimizing cone paths and then to almost minimizing admissible paths in M. Proposition 4.2 identifies the latter infimum with D. Taking both infima gives

$$(1 - C'_E\rho^2)d_{\mathcal{C}_E}(x, y) \le D(\Psi_E x, \Psi_E y) \le (1 + C'_E\rho^2)d_{\mathcal{C}_E}(x, y) \quad (r(x), r(y) \le \rho). \tag{31}$$

The slightly larger radius used by the competitors is absorbed into C'_E. Setting $y = 0$ proves (7).

For completeness, fix $L < \infty$ and put $x = \epsilon\hat{x}$, $y = \epsilon\hat{y}$ in (31). On the cone ball of radius L, the map $\hat{x} \mapsto \Psi_E(\epsilon\hat{x})$ has distance distortion $O_{E,L}(\epsilon^2)$ when the target distance is divided by ϵ. The radial formula shows that the preimage of the closed target unit ball lies between cone radii $1 - O_E(\epsilon^2)$ and $1 + O_E(\epsilon^2)$. Radial adjustment to radius one costs $O_E(\epsilon^2)$ in the rescaled metric. This gives a correspondence of the two closed unit balls with that distortion, hence the Gromov–Hausdorff estimate in Corollary 1.2. The construction works for every sequence $\epsilon \downarrow 0$; the metric tangent cone is consequently unique.

5. The determinant class and the canonical metric

Write $R = 2r_0$, $c = 2aH$ and $h = c_1(H) = [\omega]$. Let Ω be the Kähler form of the harmonic metric on M_{reg}. In the normalization of Biswas–Schumacher, denote the Weil–Petersson form by ω^{WP}. There is a fixed positive constant b such that

$$\Omega = b\,\omega^{\mathrm{WP}}. \tag{32}$$

This constant only converts trace and real-metric conventions; the cone and actual metrics throughout the paper use the same harmonic convention. We retain b explicitly in the class calculation.

5.1. **The positive determinant polarization.** Choose a sufficiently divisible positive integer k, a divisor $D \in |H^k|$, and a point $x \in S$. For the fixed Mukai vector $v = 2v_0$ define

$$u_D = -R[\mathcal{O}_D] + \chi(F \otimes \mathcal{O}_D)[\mathcal{O}_x], \qquad \lambda_D = \det Rp_*(\mathcal{E} \otimes q_S^* u_D). \tag{33}$$

The Euler characteristic depends only on v, not on F. The expression on the right is the determinant of cohomology on family parameter spaces, descended to M. It is the positive convention of [HL97, Definition 8.1.9]. Here p and q_S are the projections to the parameter space and S. Any positive power required in the construction can be absorbed into λ_D, with the reciprocal change in the scalar class coefficient below. The auxiliary k does not change the fixed form ω used for the harmonic metric.

The stabilizer weights also make descent explicit in this family. For a stable factor G of vector v_0, additivity of the Euler characteristic gives

$$\chi(G \otimes u_D) = -2r_0\chi(G \otimes \mathcal{O}_D) + 2\chi(G \otimes \mathcal{O}_D)r_0 = 0.$$

Each multiplicity-group determinant character is therefore trivial at a closed polystable orbit, including a repeated summand. At a stable point the same equality follows directly from the definition of u_D.

Lemma 5.1. *The determinant line λ_D is ample on M.*

Proof. The determinant construction is semiample and defines the projective Donaldson–Uhlenbeck morphism [HL97, Definition and Theorem 8.2.8]. Its closed-point invariant is the reflexive hull of the slope graded object together with the defect cycle of that graded object [HL97, Theorem 8.2.11]. Proposition 2.1 says that every slope graded factor here is locally free. Thus the defect cycle is zero. The remaining invariant is the polystable bundle itself, and slope S-equivalence agrees with Gieseker S-equivalence. The morphism is therefore injective on closed points. Being projective and quasi-finite, it is finite. A positive power of λ_D is the pullback of an ample line under this finite morphism, proving ampleness. This argument needs neither normality of the target nor an isomorphism with it. □

Lemma 5.2. *With the determinant convention (33), before replacing it by a descent power,*

$$[\Omega] = \frac{4\pi^2 b}{Rk}\, c_1(\lambda_D) \quad \text{in } H^2(M_{\mathrm{reg}}, \mathbb{R}). \tag{34}$$

Proof. The Chern character of the K-theory class in (33) is

$$\mathrm{ch}(u_D) = (0, -Rkh, k(c\cdot h)[x]). \tag{35}$$

Indeed, $\mathrm{ch}(\mathcal{O}_D) = 1 - e^{-kh}$ and $\chi(F \otimes \mathcal{O}_D) = k(c\cdot h) - Rk^2h^2/2$ on a K3 surface.

Work first on a stable parameter space with a universal family $\mathcal{E}$. Since $H^1(S, \mathbb{R}) = 0$,

$$c_1(\mathcal{E}) = q_S^* c + p^*\beta$$

for a class β on that space. Grothendieck–Riemann–Roch and (35) give

$$\begin{aligned} c_1(\lambda_D) &= -Rk\, p_*(\mathrm{ch}_2(\mathcal{E})\, q_S^* h) + k(c\cdot h)\beta \\ &= -\frac{k}{2}\, p_*(\mathrm{ch}_2(\mathrm{End}\,\mathcal{E})\, q_S^* h). \end{aligned} \tag{36}$$

For the second equality use $\mathrm{ch}_2(\mathrm{End}\,\mathcal{E}) = 2R\,\mathrm{ch}_2(\mathcal{E}) - c_1(\mathcal{E})^2$ and $p_*(c_1(\mathcal{E})^2 q_S^* h) = 2(c\cdot h)\beta$. This agrees with the discriminant formula in [HL97, Proposition 8.3.1(ii)].

The fixed-determinant curvature identity of [BS16, Proposition 4.3, equation (4.6)], in surface dimension, is

$$\frac{\omega^{\mathrm{WP}}}{2\pi^2} = -\frac{1}{R}\, p_*(\mathrm{ch}_2(\mathrm{End}\,\mathcal{E}, h_E) \wedge \omega). \tag{37}$$

The fixed-determinant hypothesis holds because $\mathrm{Pic}^0(S) = 0$. Comparing (36) and (37) gives the stated coefficient.

One can check the sign through the Quillen line as well. In the convention of [BS16, Theorem 4.6, equations (4.11) and (4.13)], the positive line is the dual determinant

$$Q_D = \left[\det Rp_*\left((\mathrm{End}\,\mathcal{E} - R^2\mathcal{O}) \otimes (q_S^* H^k - q_S^* H^{-k})\right)\right]^{-1}. \tag{38}$$

Its first Chern class is

$$c_1(Q_D) = -2k\, p_*(\mathrm{ch}_2(\mathrm{End}\,\mathcal{E})\, q_S^* h) = 4c_1(\lambda_D).$$

Using the nondual determinant would reverse this sign.

Finally these computations give equality on the coarse stable locus, not merely on a family. The stable Quot parameter space is a principal $\mathrm{PGL}(m)$ bundle over M^s for an appropriate framing dimension m. Its universal endomorphism bundle and determinant classes descend. Pullback in real degree-two cohomology is injective, by the low-degree Leray sequence and $H^1(\mathrm{PGL}(m), \mathbb{R}) = 0$. Thus the equality proved on that atlas gives (34) on $M^s = M_{\mathrm{reg}}$. No universal bundle on the full coarse moduli space is required. □

Put $\lambda = \lambda_D$ and $\kappa = 4\pi^2 b/(Rk)$, adjusted if a positive descent power has been taken. Choose a smooth Kähler representative $\theta \in c_1(\lambda)$, obtained from a projective embedding and scaled by its degree.

5.2. **Continuous potentials in the global class.** Theorem 11 of [BSc21] applies to polystable bundles deformable to stable bundles over a compact Hodge manifold. The surface S is Hodge since $[\omega] = c_1(H)$; all our polystable objects are bundles; and the simple points in Section 2.2 show local density of stable bundles. The analytic moduli germs are those identified in Section 3. That theorem therefore gives continuous plurisubharmonic local potentials for Ω. We now check that they define a current in the global class from Lemma 5.2.

The variety M is Cohen–Macaulay and $\operatorname{codim} \operatorname{Sing} M \geq 6$. Scheja's restriction theorem [Sch61, Section 5, Corollary after Satz III] gives

$$H^1(M, \mathcal{O}_M) \simeq H^1(M_{\mathrm{reg}}, \mathcal{O}_{M_{\mathrm{reg}}}). \tag{39}$$

In the theorem's depth convention, for degree one it suffices that $\operatorname{depth}_x \mathcal{O}_M \geq \dim_x \operatorname{Sing} M + 3$; our dimension and codimension bounds satisfy this. Combining (39) with (13) gives vanishing of the right side.

By Lemma 5.2, there is a smooth real one-form α on M_{reg} with $d\alpha = \Omega - \kappa\theta$. Its $(0,1)$ part is $\bar\partial$-closed. The preceding vanishing and the Dolbeault theorem give a smooth function f with $\alpha^{0,1} = \bar\partial f$. Since α is real,

$$\Omega = \kappa\theta + dd^c\varphi \quad \text{on } M_{\mathrm{reg}}, \qquad \varphi = 2\operatorname{Im} f. \tag{40}$$

Let U be a small neighbourhood in M. Write $\Omega = dd^c u$ on U_{reg}, where u is the continuous local potential supplied above, and $\theta = dd^c w$ with w a smooth local potential. Then

$$h_U = \varphi - u + \kappa w$$

is pluriharmonic on U_{reg}. Both h_U and $-h_U$ are plurisubharmonic there. The Grauert–Remmert extension theorem across a subset of complex codimension at least two in a normal space [GR56, Section 1.7, Satz 4] extends both as plurisubharmonic functions on U. In this codimension the theorem requires no prior upper bound. Their sum is zero on the dense regular locus, hence everywhere by uniqueness of plurisubharmonic extension. The extensions are consequently finite and continuous: each is both upper and lower semicontinuous. Their dd^c currents are opposite positive currents, so both vanish.

It follows that $\varphi = u - \kappa w + h_U$ extends continuously over U. The extensions agree on overlaps by density, producing a global continuous function φ on M. Formula (40) now defines a positive closed current on M in $\kappa c_1(\lambda)$, with the claimed local continuous potentials.

5.3. **Identification by Ricci-flat uniqueness.** The harmonic metric on the stable locus is hyperkähler. This is Itoh's theorem for irreducible Einstein–Hermitian connections on a K3 surface [Ito88, Theorem 2.1 and Remark (iii)]; the Einstein–Hermitian extension in that remark allows the nonzero slope here. The associated holomorphic symplectic form, in harmonic coordinates, is a fixed nonzero multiple of

$$(a, b) \longmapsto \int_S \operatorname{tr}(a \wedge b) \wedge \sigma. \tag{41}$$

Indeed, the quaternionic complex structures on endomorphism-valued one-forms and the trace L^2 pairing give (41) by contraction with the parallel form σ. This is the Serre-duality form defining the Mukai symplectic structure, so it agrees with σ_M up to the fixed normalization.

This identification is a calculation of the forms; it does not require extending a Riemannian tensor across Σ.

Set $d = \dim_{\mathbb{C}} M = 4q + 2$. The hyperkähler volume identity on M_{reg} says that Ω^d is a constant positive multiple of the adapted measure defined by $\sigma_M^{d/2}$ and its conjugate. The constant is determined by the total volume of $\kappa c_1(\lambda)$. The current has locally bounded potentials, so its Monge–Ampère measure does not charge the singular analytic set. The volume identity therefore holds on all of M.

The variety is projective, has canonical singularities and trivial canonical bundle. The class $\kappa c_1(\lambda)$ is Kähler and belongs to the real Néron–Severi space. The existence and uniqueness theorem of Eyssidieux–Guedj–Zeriahi [EGZ09, Theorem 7.5 and Corollary 7.6] applies to precisely these data. It identifies our current with the canonical singular Ricci-flat current in that class. This proves part (i) of Theorem 1.1. Together with Section 4, it proves all claims of Corollary 1.2 for the canonical metric.

6. The metric on the singular strata

We finish by distinguishing the quadratic approximation from an exact local cone metric. The argument uses only the intrinsic recognition of the two singular strata and their harmonic metrics.

Lemma 6.1. *The subset Σ consists precisely of the points of M whose metric tangent cones are not Euclidean. Within Σ, the subset Δ is its topological nonmanifold locus. The corresponding descriptions hold in the cones (14) and (15).*

Proof. At a regular point the tangent cone is Euclidean. At an off-diagonal point the normal cone is $\mathcal{N}_q = T^*\mathbb{C}^q /\!\!/\!\!/_0 \mathrm{U}(1)$. Its link is

$$L_q = V_2(\mathbb{C}^q)/S^1, \tag{42}$$

where $V_2(\mathbb{C}^q)$ is the complex Stiefel manifold of orthonormal two-frames and the circle multiplies both vectors by the same phase. To see this, conjugate the cotangent coordinate in the equations $y(x) = 0$, $|x| = |y|$ and then fix the common norm.

The fibration $S^{2q-3} \to V_2(\mathbb{C}^q) \to S^{2q-1}$ gives $\pi_1(V_2) = \pi_2(V_2) = 0$ for $q \geq 4$. The circle bundle over L_q consequently gives $\pi_1(L_q) = 0$ and $\pi_2(L_q) = \mathbb{Z}$. Hurewicz then gives $H_2(L_q, \mathbb{Z}) = \mathbb{Z}$. Thus the vertex of $\mathcal{N}_q$ has nonzero local homology in degree three, strictly below its real dimension $4q - 4$. Multiplication by the Euclidean factor shifts this nonzero local homology but does not remove it. The off-diagonal cone is therefore not a topological manifold at its vertex, and in particular is not Euclidean.

The diagonal cone contains off-diagonal points arbitrarily close to its vertex: choose two distinct one-dimensional loop characters tending to zero and take their direct sum. At each such point the normal slice has the off-diagonal form just considered. Hence the diagonal cone also cannot be Euclidean. Uniqueness of tangent cones from Section 4 proves the first assertion on M. The finite-dimensional slice theorem gives the same assertion at all points of each cone.

By (12), $\Sigma = \mathrm{Sym}^2 N$. Off the diagonal it is a manifold. Near a diagonal point its germ is

$$\mathbb{C}^n \times (\mathbb{C}^n/\{\pm 1\}). \tag{43}$$

The normal link is $\mathbb{RP}^{2n-1}$, whose first integral homology is $\mathbb{Z}/2$. This gives nontop local homology at the vertex, so the nonmanifold locus of Σ is precisely Δ. In the diagonal cone, the nonsimple semisimple representations are unordered pairs of one-dimensional loop characters. Their space is $\mathrm{Sym}^2 \mathbb{C}^n$, with the same description (43); its nonmanifold locus is the diagonal

flat factor $\mathbb{C}^n$. In the off-diagonal cone the singular set is the flat factor $\mathbb{C}^{2n} \times \{0\}$ and is everywhere a manifold. This proves all the claimed intrinsic descriptions. □

Lemma 6.2. *Let S_0 be either Δ or $\Sigma \setminus \Delta$. The intrinsic length metric obtained by restricting curves to S_0 and measuring their lengths with the completed ambient distance is the Riemannian length metric of its harmonic stratum metric. The same assertion holds for the corresponding flat strata in the tangent cones.*

Proof. Fix $s \in S_0$. In the chart centred at s, the stratum is the fixed factor $U \times \{0\}$. The coarse tensor bounds and the exit estimate make the restricted ambient distance locally bi-Lipschitz equivalent to a smooth metric on S_0. Thus a rectifiable curve has a locally absolutely continuous parametrization in its smooth coordinates.

For a differentiable curve γ with $\gamma(0) = s$, write its fixed-factor coordinate as $u(t) = t\dot{u}(0) + o(t)$. The radius formula (7) in this fixed centred chart gives

$$\lim_{t \to 0} \frac{D(\gamma(t), \gamma(0))}{|t|} = \|\dot{u}(0)\|_U = \|\dot{\gamma}(0)\|_{g_{S_0}}. \tag{44}$$

The last equality follows from the harmonic normalization at the centre. The argument also applies at almost every differentiability point of an absolutely continuous curve. It is pointwise: the comparison constant belongs to the fixed centre at which the limit is taken.

The length of a Lipschitz curve in a metric space is the integral of its metric speed [BBI01, Theorem 2.7.6]. Applying this locally, and subdividing a rectifiable curve, proves equality of its ambient-metric length and harmonic Riemannian length. Infima over curves in the stratum prove the assertion. For the finite cones the same proof uses the finite metric slice and its radius comparison. □

Proposition 6.3. *For every singular point $p \in M$, no pointed neighbourhood of (M, D, p) is isometric to a neighbourhood of the vertex in its metric tangent cone.*

Proof. An isometry preserves metric tangent cones, and hence preserves Σ by Lemma 6.1. If p is off the diagonal, a sufficiently small neighbourhood meets only the smooth stratum $\Sigma \setminus \Delta$. In the tangent cone this stratum is the flat factor $\mathbb{C}^{2n}$. If p is diagonal, an isometry also preserves the topological nonmanifold locus of Σ, so it sends Δ to the flat factor $\mathbb{C}^n$ of the diagonal cone. Lemma 6.2 shows that in either case the restricted map would be an isometry of the intrinsic Riemannian metrics on open subsets of these strata.

On the local ordered cover of $\Sigma \setminus \Delta$, the harmonic metric is $g_N \oplus g_N$. On Δ it is $2g_N$, since the tangent inclusion is $a \mapsto a \otimes I_2$ and $\operatorname{tr}(I_2) = 2$. An isometry to a flat factor therefore makes g_N flat on a nonempty open subset of N.

The metric g_N is hyperkähler, hence Ricci-flat and real analytic in harmonic coordinates [DK81, Theorem 5.2]. Analytic continuation of its curvature on the connected manifold N would make it flat everywhere. A complete simply connected flat manifold is Euclidean, whereas N is compact, simply connected and of positive dimension. This contradiction proves the proposition. □

The proposition concerns exact isometry. It provides no lower bound on the first nonzero error term in (6). Optimality of the exponent, derivative estimates near lower strata, and extensions to moduli with a singular-sheaf boundary remain separate questions.

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